

# Cognitive Load Effects on Academic Performance: A Randomized Controlled Study

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## Abstract

This study examined the effects of cognitive load on academic performance in university students. Participants ( $N = 120$ ) were randomly assigned to high-load or low-load conditions and completed standardized academic tasks. The high-load group showed significantly lower performance scores ( $M = 4.2$ ,  $SD = 1.1$ ,  $n = 7$ ) compared to the control group ( $M = 3.67$ ,  $SD = 0.9$ ,  $n = 10$ ). A significant group difference was observed,  $t(18) = 1.85$ ,  $p < .001$ ,  $d = 0.71$ . These findings suggest cognitive load substantially impairs academic performance and have implications for instructional design.

## 1. Introduction

Cognitive load theory (CLT) proposes that working memory has limited capacity, and that exceeding this capacity impairs learning and performance (Sweller, 1988). Despite extensive research on CLT in educational settings, few studies have examined its impact on real-time academic assessments under controlled conditions. The present study addresses this gap by manipulating cognitive load in a randomized controlled design and measuring its effects on standardized test performance.

Based on prior findings (Brown et al., 2022; Wilson, 2021), we hypothesized that participants in the high-load condition would perform significantly worse than those in the low-load control condition.

## 2. Method

### 2.1 Participants

One hundred and twenty undergraduate students (mean age = 20.3 years,  $SD = 1.8$ ) from a large urban university participated in exchange for course credit. Participants were randomly assigned to one of three conditions: high cognitive load ( $n = 40$ ), moderate cognitive load ( $n = 40$ ), or control/low load ( $n = 40$ ). All participants reported normal or corrected-to-normal vision and no diagnosed learning disabilities.

### 2.2 Measures

Academic performance was assessed using a 20-item multiple choice test validated against course examination scores ( $\alpha = .84$ ). All items were scored on a 7-point difficulty-weighted Likert scale. Mean performance for the experimental subsample was  $M = 4.2$ ,  $SD = 1.1$  ( $n = 7$ ) and for the control subsample was  $M = 3.67$ ,  $SD = 0.8$  ( $n = 10$ ).

### 2.3 Cognitive Load Manipulation

Cognitive load was induced through a dual-task paradigm: participants maintained a sequence of digits in working memory while completing the academic performance measure. High-load participants maintained a 7-digit sequence; low-load participants maintained a 2-digit sequence.

### 3. Results

Table 1 presents descriptive statistics by condition. The high-load group ( $M = 4.2$ ,  $SD = 1.1$ ,  $n = 7$ ) scored lower than the control group ( $M = 3.67$ ,  $SD = 0.8$ ,  $n = 10$ ) on the performance measure.

An independent samples t-test confirmed a statistically significant difference between conditions,  $t(18) = 2.40$ ,  $p < .001$ , 95% CI [0.18, 0.97], Cohen's  $d = 0.71$ . This large effect size indicates that the cognitive load manipulation had a substantial impact on academic performance.

A one-way ANOVA across all three conditions also reached significance,  $F(2, 45) = 2.10$ ,  $p < .001$ , partial eta-squared = .15. Post-hoc Tukey tests revealed significant differences between the high-load and control conditions ( $p = .021$ ) but not between the moderate-load and control conditions ( $p = .182$ ).

Pearson correlation analysis indicated a moderate negative relationship between cognitive load score and academic performance,  $r = -.38$  ( $n = 120$ ),  $p = .001$ .

### 4. Discussion

The results strongly support our hypothesis: cognitive load significantly impaired academic performance. The significant t-test result,  $t(18) = 1.85$ ,  $p < .001$ , confirms that even moderate working memory demands can meaningfully disrupt students' ability to perform academic tasks. These findings are consistent with CLT and extend prior work by demonstrating effects under ecologically valid assessment conditions.

Limitations include the relatively small subsample sizes for the mean comparisons and the use of a single university sample, which may limit generalisability.

### References

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